POWER ELECTRONICS AND DRIVES

LABORATORY MANUAL



DEPARTMENT OF ELECTRICAL ENGINEERING

COLLEGE OF ENGINEERING TRIVANDRUM

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Chapter 5

Effect of source inductance in single-phase controlled rectifier feeding inductive load

5.1 Objective

Aim: To set up a single-phase fully controlled rectifier with source inductance supplying highly inductive load. Observe relevant waveforms and the average voltage lost due to the effect of source inductance. Simulate the circuit using any suitable simulation software and find the voltages, input power factor, line current and THD. Compare the results.

5.2 Theory

The full controlled rectifier has four thyristors. At a time two thyristors, one from top and the other from bottom will conduct. During positive half cycles of input voltage, thyristors T1 and T3 are triggered at $\omega t = \alpha$; similarly, T2 and T4 are triggered at $\omega t = \pi + \alpha$. For a full converter supplying inductive load, the output is present during positive and negative half cycles. The average output voltage can be either positive or negative which depends on firing angle.

In a single-phase rectifier, source inductance (L_s) significantly impacts performance by preventing the instantaneous commutation of current between devices. Its main effects are:
Commutation Overlap: Because an inductor resists sudden changes in current, the current in the outgoing thyristors cannot drop to zero immediately, nor can it rise to full value in the incoming thyristors instantaneously.
Overlap Angle (μ): During commutation, all devices (incoming and outgoing) conduct simultaneously for a period known as the overlap interval, measured by the overlap angle μ.

Reduction in Output Voltage: During the overlap period, the AC source is effectively

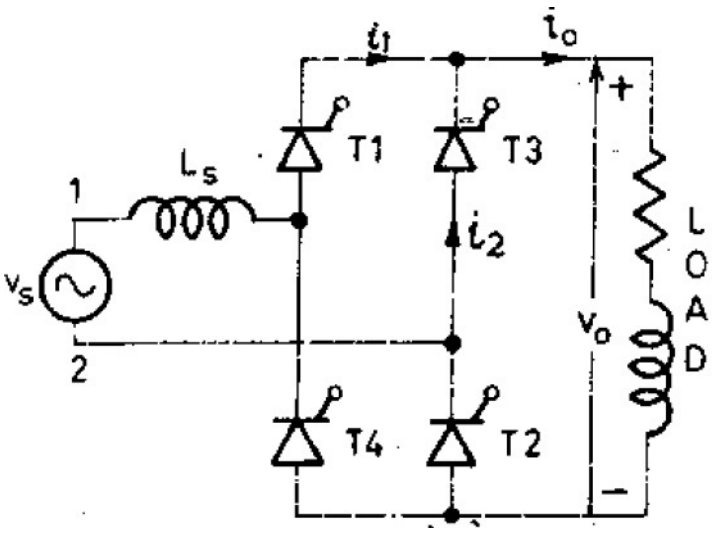


Figure 5.1: Power circuit of full converter with source inductance supplying R-L load.

short-circuited through the conducting thyristors, causing the output voltage to drop to zero (in single-phase bridge rectifiers) or to the average of the phase voltages.
Average DC Voltage Drop: The presence of source inductance results in a reduction of the average output voltage (V_{dc}) proportional to the load current (I_0) and the source inductance (L_s).

The reduction in average output voltage (ΔV_o) and the final average output voltage (V_o) for a single-phase fully controlled bridge rectifier are given by:
Average Voltage Reduction: $\Delta V_o = \frac{\omega L_s I_0}{\pi} = 4 f L_s I_o$
Average Output Voltage: $V_o = \frac{2V_m}{\pi}\cos(\alpha) - \frac{\omega L_s I_0}{\pi}$
Overlap Angle Relation: $V_m(\cos\alpha - \cos(\alpha+\mu)) = 2\omega L_s I_0$

Power Factor: The delay in commutation causes the input current to lag further behind the voltage, thereby reducing the input power factor.
Commutation Failure: At high firing angles ($\alpha \approx 180°$), the overlap may exceed the available turn-off time, leading to a failure in the inverting mode of operation.
Harmonics: The distorted current waveforms during overlap introduce higher-order harmonics into the AC source.

Figure 5.2 shows the voltage across different devices and the dc output voltage during each of these conduction modes.

Under normal operating condition of the converter the load current may or may not remain zero over some interval of the input voltage cycle. If i_0 is always greater than zero then the converter is said to be operating in the continuous conduction mode. In this mode of operation of the converter $T_1 - T_2$ and $T_3 - T_4$ conducts for alternate half cycle of the input

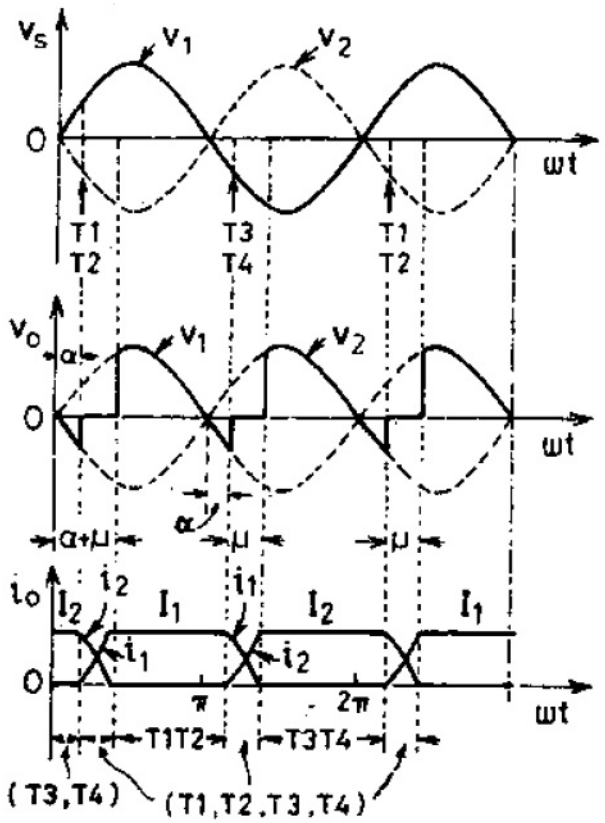


Figure 5.2: Output wave forms of full bridge converter with source inductance supplying R-L load

supply. However, in the discontinuous conduction mode none of the thyristors conduct over some portion of the input cycle. The load current remains zero during that period.

Sl.No.	$\alpha, FiringAngle()$	Output voltage without L_s	Output voltage with L_s

5.3 Procedure

Set up the circuit. Connect the power supply to the rectifier through isolation transformer to avoid any ground looping. Observe the waveforms without source inductance L_s for RL load. Vary firing angle and observe the wave forms across the load and thyristors. Repeat the experiment after connecting the source inductance L_s. Repeat the earlier procedure and obseve the output variation with firing angle variation. Compare the average out put voltage obtainced for the two cases.

5.4 Result

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5.5 Simulation Result

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5.6 Inference

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5.7 Outcome

After doing this experiment, the student will be able to understand the effect of source inductance at the controlled rectifier output and input parameters.

5.8 Questions

- What is the difference in current and voltage wave forms for controlled rectifier with and without source inductance?

Chapter 6

Measurement of reverse recovery current and reverse recovery time

6.1 Objectives

Aim: To measure the peak reverse current, reverse recovery time and compute the softness factor and power losses of a high frequency diode. Compare it with the theoretical estimate from data sheet and that of other diodes. Simulate the circuit using any suitable simulation software and compare the results.

6.2 Theory

For the diode to transit from the conducting to the reverse blocking state, the charge distribution must change. This can only happen with a movement of charge from the junction, which is a flow of current. The diode temporarily conducts in the reverse direction. The maximum current (I_{RRM}) is the reverse recovery current, and is specified in the diode datasheet. It increases greatly with temperature. It increases with di/dt. Time interval t_a is defined as the time duration between the instant of zero crossing and reaching the peak of the reverse current. Time interval t_b is defined as the time between the peak of the reverse current and the time when the current falls to 25% . The sum of t_a and t_b is called the reverse recovery time, t_{RR}. Using a triangular approximation for the current waveform shown in Figure 6.1:

$$Q_{rr} \approx \frac{1}{2} \cdot I_{rr} \cdot t_{rr}$$

Where:Q_{rr} = Reverse recovery charge (Coulombs).I_{rr} = Peak reverse recovery current (Amperes).

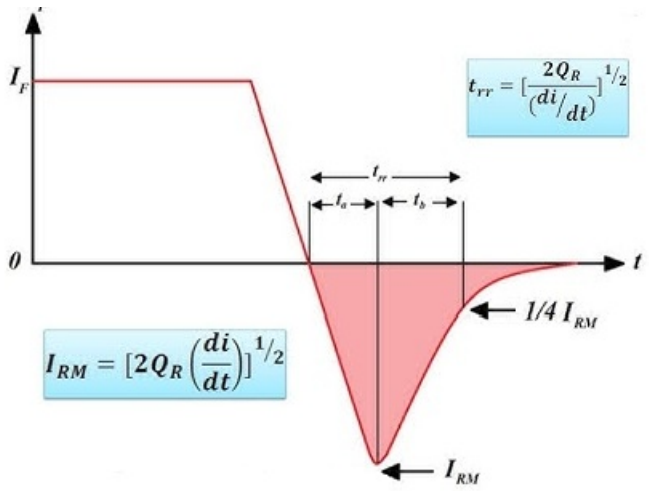


Figure 6.1: Reverse recovery charactristics of high frequency diode

Relationship with di/dt.
In practical circuits, I_{rr} depends on the rate of change of current (di/dt) imposed by the external circuit:

$$I_{rr} = \sqrt{2 \cdot Q_{rr} \cdot \frac{di}{dt}}$$

Consequently, the recovery time can be expressed as:

$$t_{rr} = \sqrt{\frac{2 \cdot Q_{rr}}{di/dt}}$$

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Power Loss in a Diode (P_{loss})
Total power dissipation in a diode is the sum of Conduction Loss and Switching Loss.
Conduction Loss (P_{cond}) is the loss while the diode is in the "ON" state. It depends on the forward voltage drop (V_F) and the current (I_F). For a constant current with duty cycle D:

$$P_{cond} = V_F \cdot I_{avg} = V_F \cdot I_F \cdot D$$

For more precise modeling (using the threshold voltage V_T and on-resistance R_d):

$$P_{cond} = (V_T \cdot I_{F,avg}) + (R_d \cdot I^2_{F,rms})$$

Switching Loss (P_{sw})Switching loss occurs primarily during the reverse recovery phase. It is proportional to the reverse voltage (V_R) across the diode and the switching frequency (f_{sw}).The energy lost during a single reverse recovery event (E_{rr}) is approximately:

$$E_{rr} \approx \frac{1}{2} \cdot V_R \cdot Q_{rr}$$

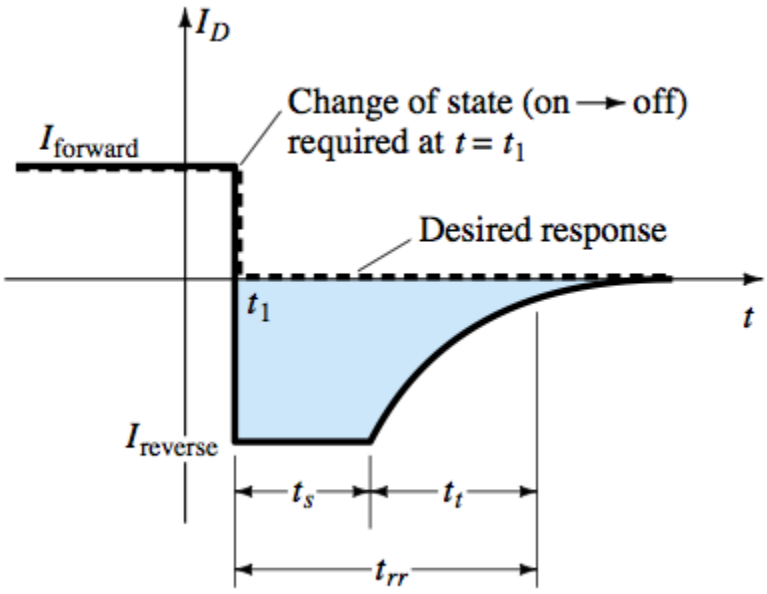


Figure 6.2: Reverse recovery charactristics of power frequency diode

(Note: Some literature uses a factor of 1/4 or 1/6 depending on the "softness" of the recovery curve.)The total switching power loss is:

$$P_{sw} = E_{rr} \cdot f_{sw} \approx \frac{1}{2} \cdot V_R \cdot Q_{rr} \cdot f_{sw}$$

Total Power Loss Equation

$$P_{total} = P_{cond} + P_{sw}$$

$$P_{total} = (V_F \cdot I_{F,avg}) + \left(\frac{1}{2} \cdot V_R \cdot Q_{rr} \cdot f_{sw}\right)$$

6.3 Procedure

A simple series circuit with a source Vs, a resistor R, and the diode under test is shown in Figure6.3. Source is a function generator providing a square wave to switch the diode rapidly between forward and reverse bias.

To find the diode's reverse recovery characteristics experimentally, use a circuit with a pulsating voltage of peak 10V from a function generator, resistor of 1kΩ, and the diode,

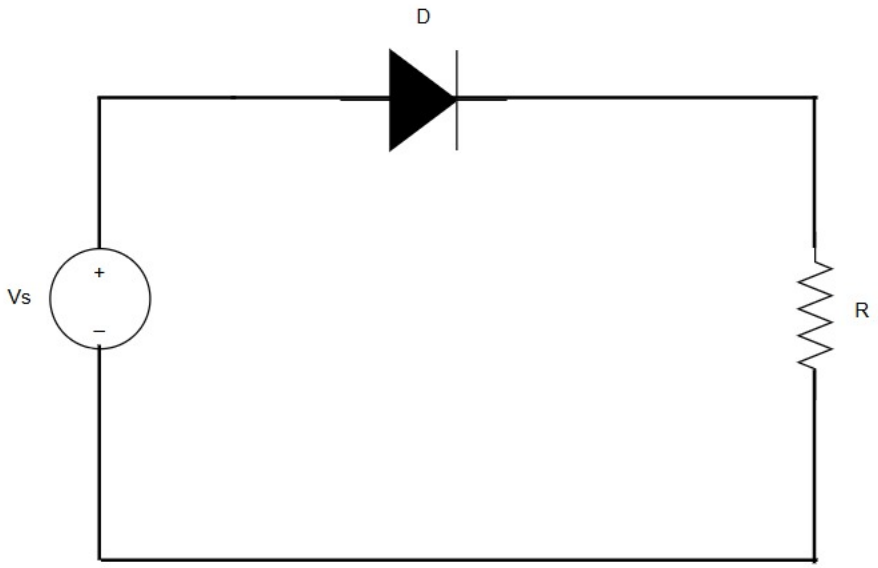


Figure 6.3: Diode circuit

driven by a fast-switching signal (like a function generator) to repeatedly forward-bias and then reverse-bias the diode, observing the resulting reverse recovery current waveform (voltage across the resistor) on an oscilloscope to measure trr (reverse recovery time), compute Irm (maximum reverse reovery current) from the measurement and identify the soft/snappy recovery by the S-factor (ratio of reverse current decay times ta and tb).

Measurement can be done with an oscilloscope visualizing the voltage across the diode and current through it (measured across the resistor R). Apply positive and negative pulses from the generator, forward and reverse biasing the diode, causing it to conduct current and block reverse voltage. When the pulse switches to negative, the diode starts to turn off from conduction. Observe current at the oscilloscope.

It will show the diode current momentarily going negative (reverse recovery current, $(t_r r)$) before settling to near zero. Measure $(t_r r)$, by measuring(ta), time from forward current zero-crossing to peak reverse current (I_{RM}) and t_b, time from (I_{RM}) to when reverse current decays to 25% of I_{RM}, (0.25 I_{RM}). Total reverse recovery time $(t_{rr} = t_a + t_b)$.
Calculate S-Factor:
$S = t_b/t_a$.
$S = 1$ (Soft Recovery). Even decay, less voltage overshoot.
$S < 1$ (Snappy/Fast Recovery): Abrupt decay, potential voltage ringing.

Key parameters to determine:
t_{rr} (Reverse recovery time) is how long it takes for the diode to stop conducting in reverse.
Q_{rr} (Reverse recovery charge) is the total charge removed during recovery, related to the area under the I_{RR} curve.
I_{RM} (Peak reverse current) is the maximum negative current during recovery.
S-Factor indicates recovery "softness" or "snappiness," crucial for power applications.

6.4 Result

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6.5 Simulation Result

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6.6 Inference

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6.7 Outcome

Student should be able to analyse switching speed and revesre current of diodes for various applications.

6.8 Questions

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Chapter 7

Switching characteristics of MOSFET

7.1 Objective

To study and understand the switching characteristics of a power MOSFET, measure rise time, fall time and compute switching losses at the switch during turn on and turn off for different switching frequencies. Estimate the ON state (conduction) loss at 75 degree celcius and switching loss of the given MOSFET at 1kHZ switching frequency.

7.2 Theory

Handling Precautions for MOSFET- The gate–source oxide layer is very thin. When touched with our body (that can have a minimum of 1–2 μ C charge the fields created can be extremely high so as to breakdown the oxide layer.

Metal Oxide Silicon Field Effect Transistors (MOSFET) are electronic devices used as a switching device in circuits. It is a voltage controlled device with three terminals such as Source(S), Gate(G), Drain(D). When voltage is applied to the gate, an electrical field is generated that changes the conductivity of the channel region to control the charge flow.

Switching times of the MOSFET are determined essentially by the times required for the gate driver to charge the internal capacitances. Since the drain current is a function of the gate-to-source voltage,the rate at which the drain current changes is dependent on the rate at which the gate-to-source capacitance is charged by the gate drive circuit. Likewise, the rate at which the drain voltage changes is a function of the rate at which the gate-to-drain capacitance is charged. The drain-to-source capacitance leads directly to switching loss, since the energy stored in this capacitance is lost during the transistor turn-on transition.

The ON-state loss for the MOSFET is given by

$$\mathrm{P}_{ON} = I_{drms}^2 R_{ds(ON)}$$

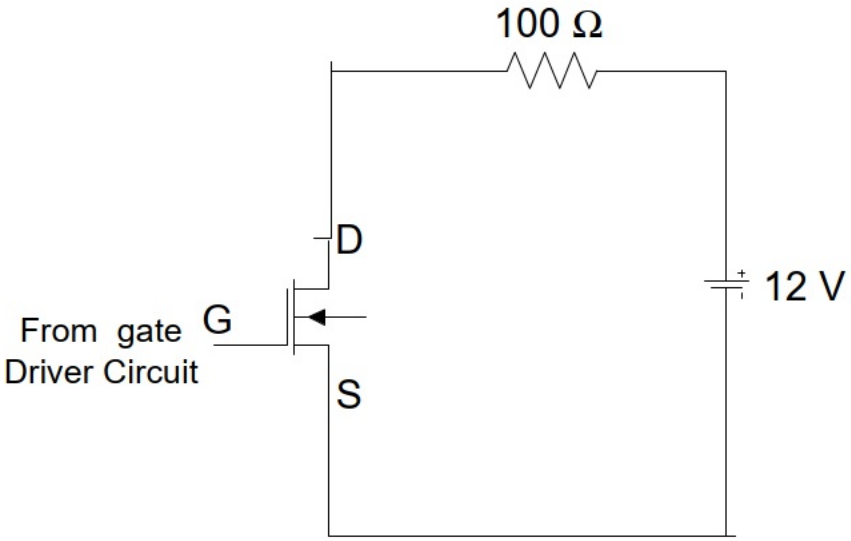


Figure 7.1: MOSFET Switching circuit

$R_{ds(ON)}$ has a temperature coefficient of approximately 0.6% per degree celcius and should be considered to calculate $R_{ds(ON)}$ and P_{ON} as the device temperature rises.

The switching loss for resistive loads, assuming linear rise and decay of V_{ds} and i_d, is given as

$$\mathrm{P}_{switching} = \tfrac{1}{6} V_{ds} I_d (t_r + t_f) f_s$$

7.3 Design

7.4 Procedure

Make connection as shown in the figure. Apply gate signal at the gate of the MOSFET through a gate driver. Measure the ON state voltage V_{ds} across the MOSFET, and load voltage using DSO accurately(use cursor option available). Compute I_d and $R_{ds(ON)}$ using the value of load resistance. Adjust the value of $R_{ds(ON)}$ for accommodating any temperature variation for finding the ON state loss.

Precisely measure the rise time t_r and fall time t_f (refer figure) during the MOSFET turn ON and OFF process (use window and cursor options in DSO). Note down the values and calculate the energy loss during switching per cycle. Compute power loss during desired switching frequency of f_s operation.

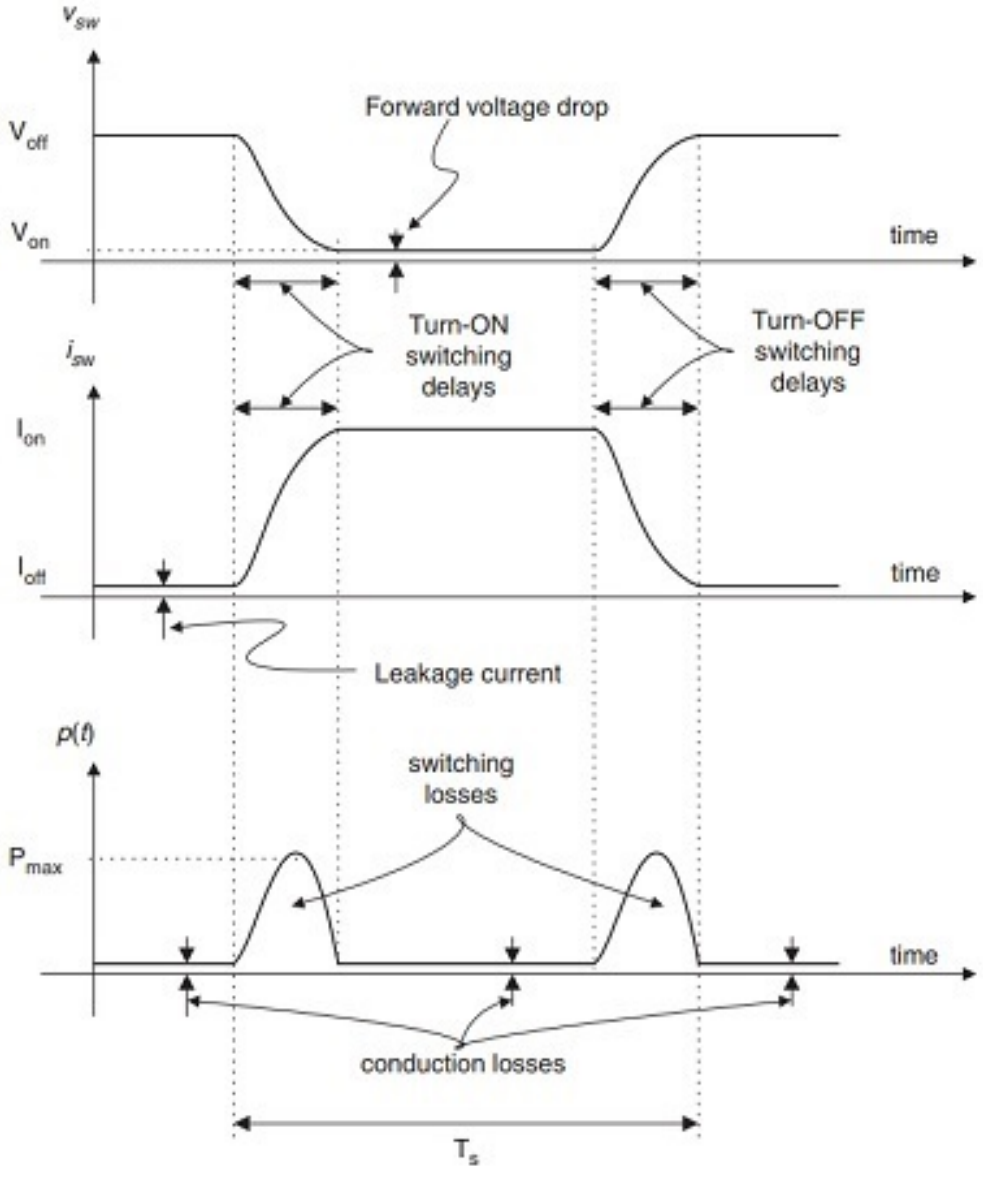


Figure 7.2: Switching loss and conduction loss in a MOSFET

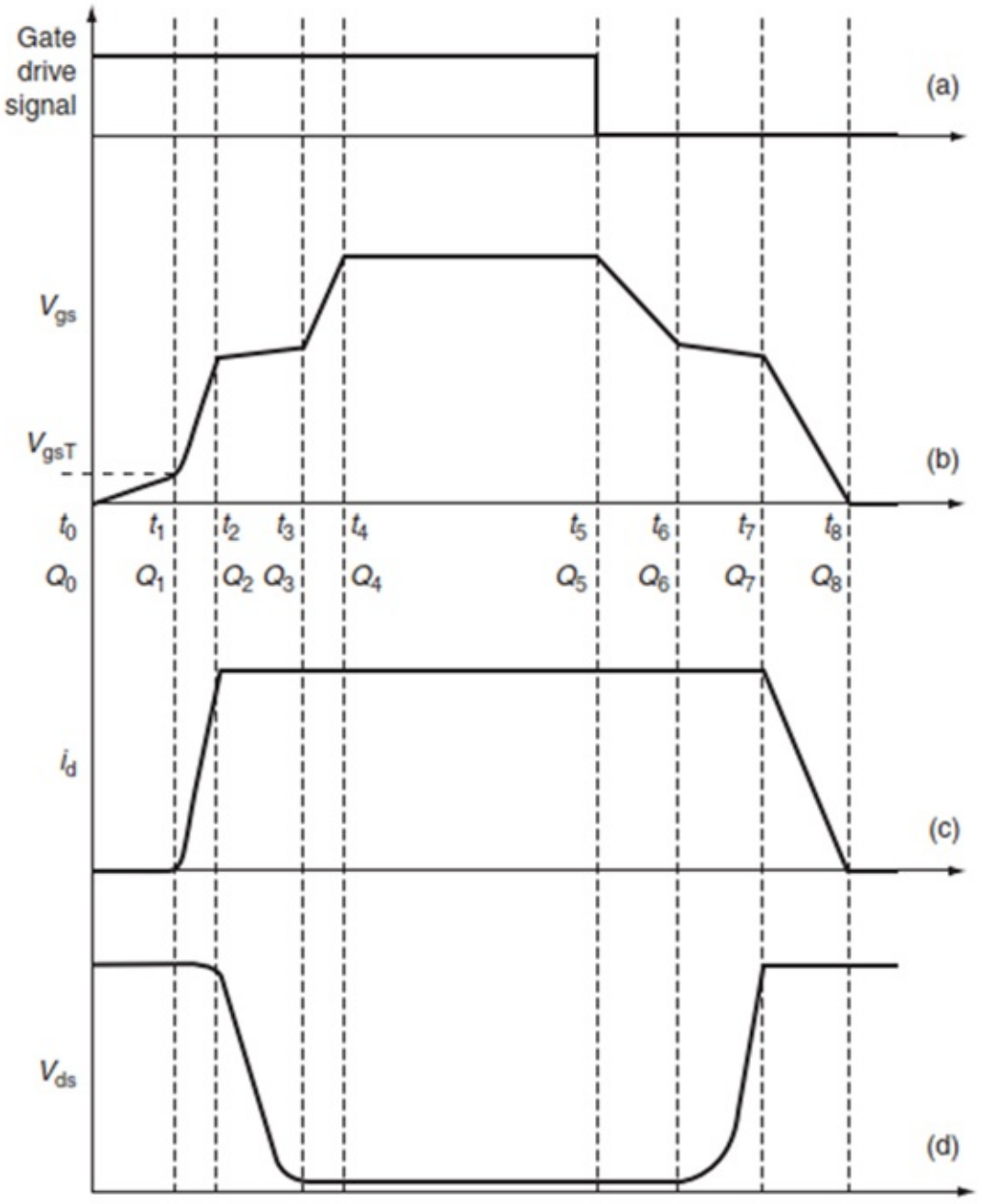


Figure 7.3: Switching characteristics of a MOSFET

7.5 Observations

Load voltage, V_{Load} =
MOSFET voltage drop, V_{ds} =
Drain current (Load current), I_d =
The ON state resistance of the MOSFET, R_{ds} =
Rise time, t_r =
Fall time, t_f =
Switching frequency, f_s =

7.6 Result

P_{ON} loss at .....=
$P_{switching}$ loss at ..... =
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7.7 Inference

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7.8 Outcome

7.9 Questions

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Chapter 8

Isolated Gate Driver Circuit for MOSFET using Optocoupler

8.1 Objective

Aim: To setup a gate driver for the MOSFET/IGBT for a certain switching frequency with galvanic isolation for low side switch/ground-referenced configurations, and to observe relevant waveforms and analyse the limitations. To simulate the gate driver circuit and compare the results.

8.2 Theory

The PC817 is one of the most widely used general-purpose optocouplers (also known as an optoisolator). It consists of an infrared emitting diode (IRED) optically coupled to a phototransistor in a compact 4-pin package. Its primary function is to provide electrical isolation between two parts of a circuit, allowing a low-power control circuit (like a microcontroller) to switch or monitor a high-power or noisy circuit without any physical electrical connection. The PC817 works through the conversion of electricity to light and back to electricity. When current flows through the internal LED (pins 1 and 2), it emits infrared light. The light travels across an internal transparent insulation layer. The light hits the base of the phototransistor (pins 3 and 4), causing it to become conductive. The amount of current the transistor allows to flow is proportional to the intensity of the light from the LED.
Pin Configuration of PC817 usually comes in a 4-pin DIP (Dual In-line Package) or SMD (Surface Mount Device) format.

Key standard specifications(exact values can vary slightly by manufacturer. Refer data sheet)
Isolation Voltage: 85000 V_{rms} (Protects against very high voltage spikes).
Current Transfer Ratio (CTR): Typically 1050% to 11600% (The ratio of output current to

input current).
Forward Current ($13I_f$): Max 50mA.
Collector-Emitter Voltage ($15V_{ce}$): Max 35V to 80V.
Response Time: Approx 3µs to 18µs (suitable for low-speed switching, not high-speed data).

Common Applications include - Microcontroller I/O Protection: Protecting delicate pins on an Arduino or STM32 from high-voltage motors or relays.
Switch Mode Power Supplies (SMPS): Providing feedback from the output side to the control side while maintaining isolation.
Noise Suppression: Preventing electrical noise from the "power side" of a circuit from reaching the "logic side.
"Level Shifting: Interfacing a 3.3V circuit with a 12V or 24V circuit.
Practical Usage -When using the PC817, a current-limiting resistor must be used in series with the input LED (Pin 1), just as with a standard LED.
To calculate the resistor (R):

$$R = \frac{V_{input} - V_{forward}}{I_{forward}}$$

(Where $V_{forward}$ is typically 1.2V and $I_{forward}$ is ideally 5mA to 20mA for longevity.)

A Low-Side Gate Driver is a specialized circuit used to drive power switches (like MOSFETs or IGBTs) where the source or emitter of the switch is connected to the common ground (the "low side"). Because the source is at 0V, driving the gate is straightforward. It simply need to apply a voltage relative to that ground to turn the switch ON. Unlike high-side drivers, low-side drivers do not require complex "bootstrap" circuits or level shifters because they are at the same ground level with the control logic. Low-side drivers are used for any circuit where the load is connected to the positive rail.

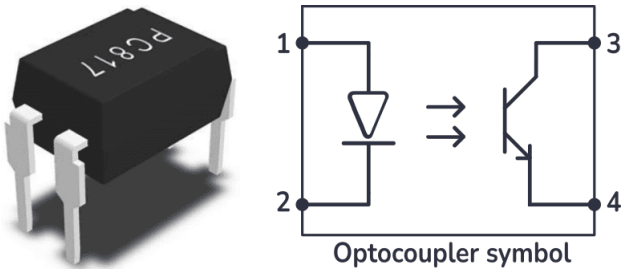


Figure 8.1: Optoisolator PC817 IC

8.3 Procedure

The circuit diagram for driving MOSFETs connected to low voltage rail using PC817 is shown in Fig. 8.2. Preferably connect an electrolytic capacitor with a value of 100uF and a

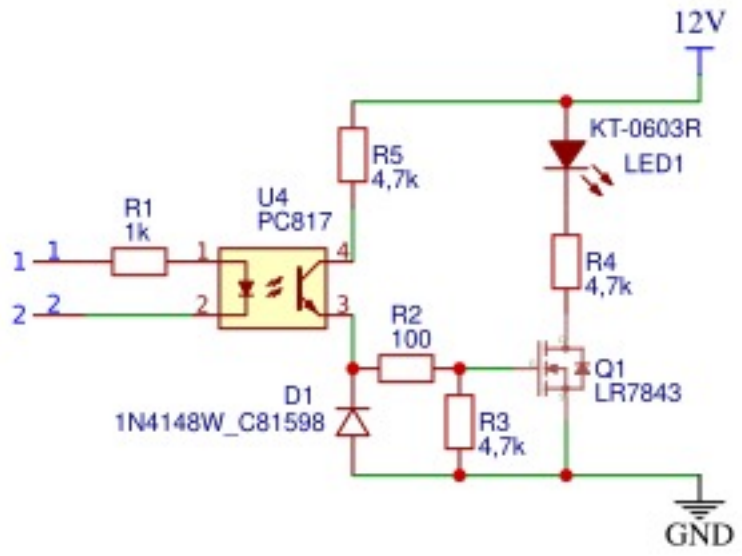


Figure 8.2: Typical low side gate driver circuit using optoisolator

ceremic capacitor of 0.1uF between the driver supply and ground (decoupling capacitors)to provide protection and a stable voltage to the IC.

The input to the IC can be a amplitude limited signal taken from a signal generator, or generated by some other means. The signal ground shoulbe isolated from other supply grounds as the initial section is an optoisolator. It must not be connected to the supply or output grounds. The load ground is to be referenced to power ground and isolated from input signal reference ground. MOSFET gate receives a high signal from PC817 when an input is high. The gate resistors are to be designed for proper charging and discharging of gate source parasitic capacitance of the MOSFET.

The power circuit voltage source and load should be selected to limit the MOSFET current to a safe value (refer data sheet of the MOSFET).

Observe and plot the gate signal and load voltage waveforms. Measure the t_{on} and t_{off} switching delays and find maximum switching frequency possible.
The thumb rule for high efficiency in power converters, the total switching transition time $(t_{on} + t_{off})$ should not generally occupy more than 1% to 5% of the total switching period (T).

$$f_{sw} \leq \frac{k}{t_{on} + t_{off}}$$

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For High Efficiency $(k = 0.01)$: Transitions take 1% of the period.
For Standard Design $(k = 0.05)$: Transitions take 5% of the period.
The absolute limit $(k = 0.1)$: Beyond 10%, the heat generated during switching usually becomes unmanageable without massive heatsinks. Remember that t_{on} and t_{off} are not fixed

constants. They depend on the Gate Resistor (R_g). For higher frequency, reduce R_g to make t_{on} and t_{off} smaller. However, smaller R_g causes higher EMI and voltage ringing.

8.4 Design, Observations and Analysis

8.5 Result

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8.6 Inference

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8.7 Outcome

After the experiment the student willbe able to design a gate driver circuit using optocoupler IC and understand the design of gate driver with isolation.

8.8 Questions

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Chapter 9

Gate Driver Circuit for high side switch

9.1 Objective

Aim:To setup a gate driver circuit for the MOSFET/IGBT for floating reference configuration. Explore bootstrapping technique and to observe relevant waveforms. Add a push pull configuration and compare the switching speed. To simulate the gate driver circuit and compare the results.

9.2 Theory

The circuit is to provide a bootstrap with push-pull gate driver to drive an N-channel MOSFET in a high-side configuration. Since the load is connected to the source, the Gate voltage must be level shifted above the 24V supply to keep the MOSFET fully ON.

Bootstrap capacitor (100nF) is the heart of the circuit. It stores energy to provide the necessary V_{GS} voltage when the MOSFET is in the high-side position.Bootstrap diode (4148) prevents the bootstrap capacitor from discharging back into the +12V supply when the MOSFET source voltage rises to 24V.

Level shifter (Bottom BC547) transistor translates the low-voltage control signal (on the left) to the higher voltage levels required by the floating bootstrap stage. Speed-up capacitor (220p) connected in parallel with the 2.2kΩ input resistor, it helps the level-shifting transistor switch faster by providing a momentary burst of current to its base.

Push-pull stage (BC547 BC557) is the complementary pair provides high peak current to quickly charge and discharge the MOSFET's gate. The BC547 (NPN) "pushes" current to turn the MOSFET on, and the BC557 (PNP) "pulls" current to turn it off. The gate resistor (10Ω) limits the peak switching current to protect the driver transistors and controls

the switching speed (t_{on} and t_{off}).

When the control signal is LOW, the level shifter is OFF. The Gate is pulled down by the BC557, turning the IRFZ44 OFF.The Source of the IRFZ44 is effectively at ground through the load.The Bootstrap capacitor (100n) charges from the +12V supply through the diode (4148). It now holds approximately 11.3V (12V minus the diode drop).

When the control signal goes HIGH, the level shifter turns ON, pulling the bases of the push-pull pair LOW.The BC547 (top) turns on. The gate of the IRFZ44 is now connected to the positive terminal of the 100nF capacitor.The MOSFET begins to conduct, and its source voltage rises toward 24V.

As the source rises to 24V, the entire bootstrap circuit "rides" on top of this voltage.The positive terminal of the capacitor is now at roughly 24V + 11.3V = **35.3**V.This provides the necessary V_{GS} (> 10V) to keep the N-channel MOSFET fully saturated even though its source is at 24V.

Design analysis and limitations of switching frequency (f_{sw})
As noted, t_{on} and t_{off} determine the maximum frequency. In this circuit, the 10Ω resistor and the current capability of the BC547/557 (approx. 100–200mA) will limit how fast the IRFZ44's gate capacitance (Q_g) can be charged.
Duty Cycle Limit: This circuit cannot stay ON indefinitely (100% duty cycle). The bootstrap capacitor will eventually discharge. The MOSFET must be turned OFF periodically to allow the capacitor to recharge from the 12V supply.

Isolation: This specific circuit shares a common ground between the control signal and the 24V power section. To achieve true galvanic isolation, an optocoupler like the PC817 would need to be placed before at the input.

9.3 Procedure

The circuit diagrams for driving MOSFETs connected to high voltage rail using bootstrap is shown in fig. 9.1. Preferably connect an electrolytic capacitor with a value of 100uF and a ceremic capacitor of 0.1uF between the driver supply and ground to provide protection and a stable voltage.

The input to the circuit can be a amplitude limited signal taken from a signal generator, or generated by some other means. The gate resistors are to be designed for proper charging and discharging of gate source capacitance of the MOSFET.

The power circuit voltage source and load should be selected to limit the MOSFET current to a safe value (refer data sheet of the MOSFET).

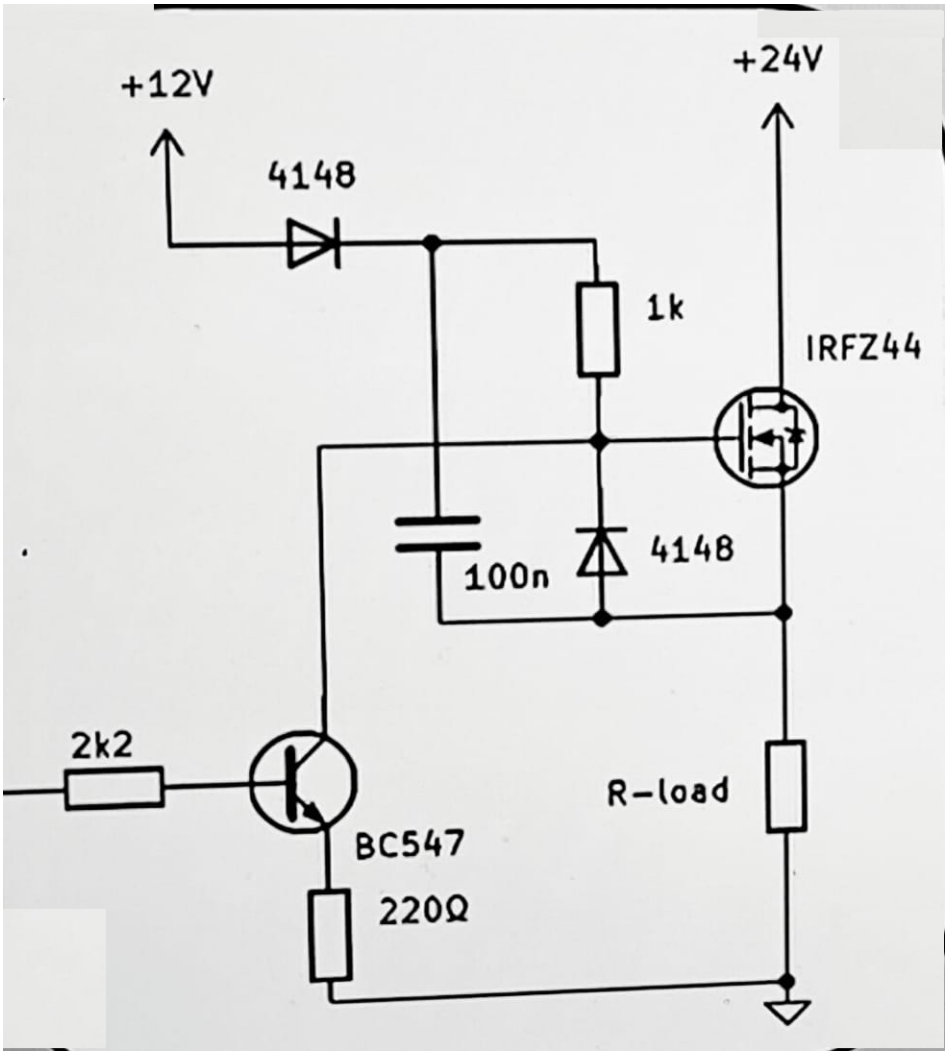


Figure 9.1: Typical boot strap circuit with push-pull output

Observe and plot the gate signal and load voltage waveforms. Measure the t_{on} and t_{off} switching delays and find maximum switching frequency possible.

Add the push pull output section of low output resistance as shown in fig.9.2.Observe and plot the gate signal and load voltage waveforms. Measure the t_{on} and t_{off} switching delays and find maximum switching frequency possible.
Compare the two cases.

The thumb rule for high efficiency in power converters, the total switching transition time $(t_{on} + t_{off})$ should not generally occupy more than 1% to 5% of the total switching period (T).

$$f_{sw} \leq \frac{k}{t_{on} + t_{off}}$$

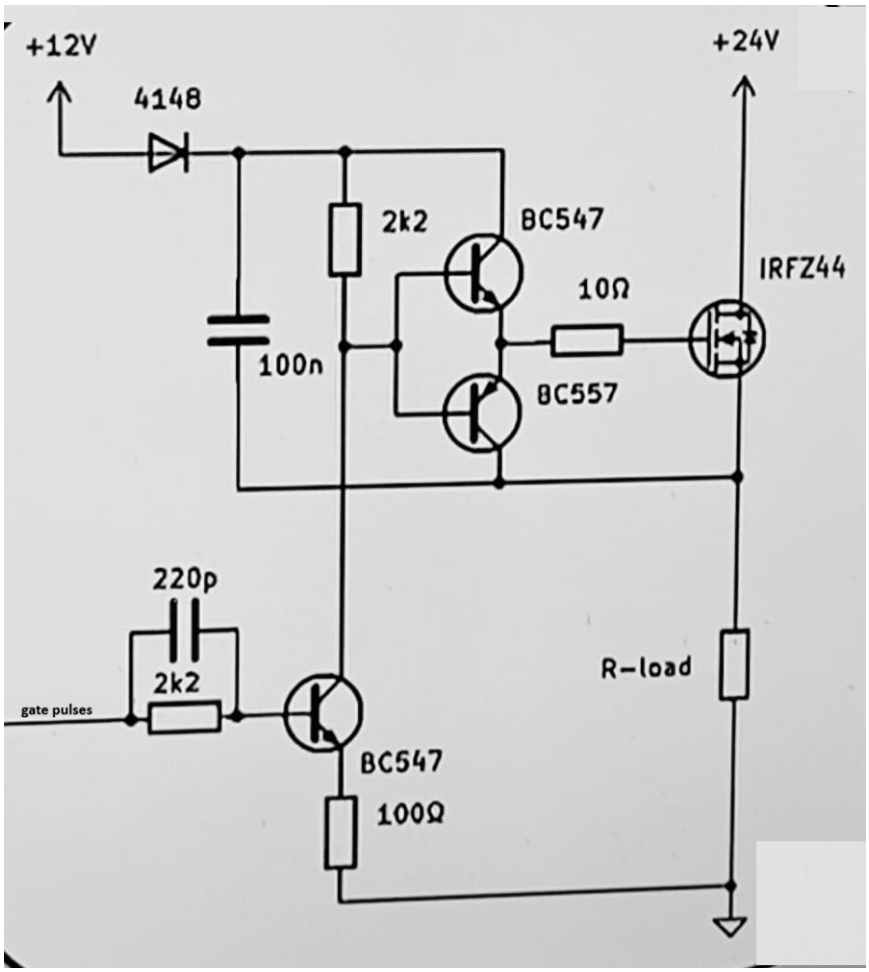


Figure 9.2: Typical boot strap circuit with push-pull output

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9.4 Design, Observations and Analysis

9.5 Result

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9.6 Inference

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9.7 Outcome

After the experiment the student willbe able to design a gate driver circuit using optocoupler IC and understand the design of gate driver with isolation.

9.8 Questions

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Chapter 10

Controlled Rectifier fed PMDC/SEDC Motor

10.1 Objective

Aim:Set up a 1-phase controlled rectifier for motor speed control. Observe the speed and torque variations also the continuous conduction waveforms. Simulate the circuit using any suitable simulation software and plot voltage, current speed, and torque waveforms.

Additional inductors may be added to make the current waveform continuous.

10.2 Theory

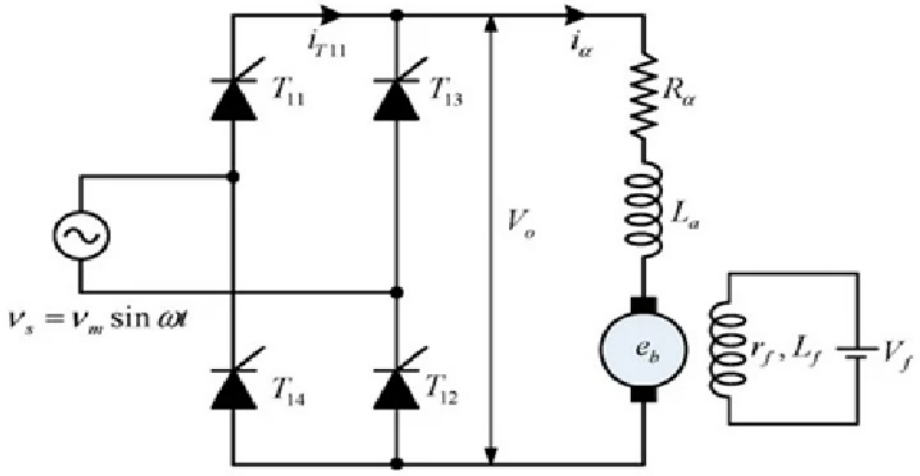


Figure 10.1: Single-phase fully-controlled converter drive

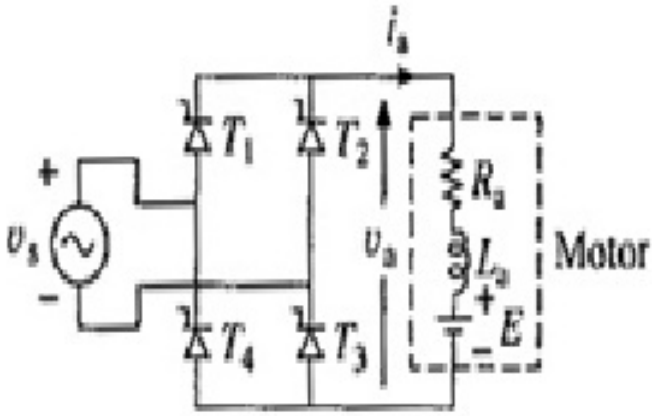


Figure 10.2: Equivalent circuit of Single-phase fully-controlled converter drive

Rectifier Fed DC Motor Drives are the standard for converting AC utility power into the variable DC voltage required for speed and torque control.

While both Permanent Magnet DC (PMDC) and Separately Excited DC (SEDC) motors share the same armature control principles, they differ fundamentally in how they handle magnetic flux.

System ArchitectureThe system consists of a phase-controlled rectifier (usually using Thyristors/SCRs) that converts AC input into a variable average DC voltage V_a by adjusting the firing angle (α).

In a separately Excited DC (SEDC) Drive motor, the field winding and the armature are powered by two independent rectifiers. Armature Rectifier controls the speed below the base speed by varying terminal voltage. Field Rectifier controls the speed above the base speed by reducing field current (Field Weakening). In permanent Magnet DC (PMDC) Drive, the field flux is provided by permanent magnets. Only one rectifier is needed for the armature. Because the magnetic field is fixed, the field weakening can not be performed. Speed control is achieved strictly by varying the armature voltage.

Principle of Operation

The speed of the motor is governed by the back-EMF equation:

$$V_a = E_b + I_a R_a$$

$$E_b = K_a \Phi \omega$$

Where V_a is the average output of the rectifier.

For a Single-Phase Fully Controlled Rectifier, the average voltage is:

$$V_a = \frac{2V_m}{\pi} \cos \alpha$$

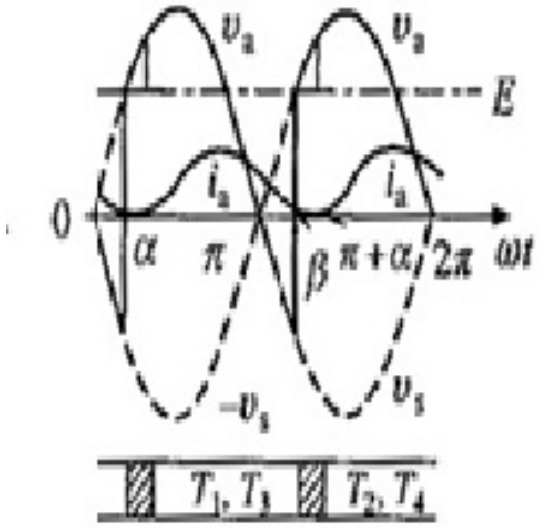


Figure 10.3: Single-phase fully-controlled converter drive output wave form in discontinuous conduction mode

Firing Angle (α) Control: As α increases (towards 90°), the average voltage V_a decreases, causing the motor speed to drop.
Conduction Modes:
Continuous Conduction: The motor inductance is high enough to keep current flowing between pulses. Speed regulation is better here.
Discontinuous Conduction: Occurs at light loads or high α. The current hits zero before the next firing pulse, leading to poor speed regulation.

The full controlled rectifier has four thyristors. At a time two thyristors, one from top and the other from bottom will conduct. During positive half cycles of input voltage, thyristors T1 and T3 are triggered at $\omega t = \alpha$; similarly, T2 and T4 are triggered at $\omega t = \pi + \alpha$. For a full converter the output is present during positive and negative half cycles. The average output voltage can be either positive or negative which depends on firing angle.

10.3 Procedure

Set up the circuit. Connect the power supply to the rectifier through isolation transformer to avoid any ground looping. Replace the load with a DC motor which can act as a RLE load. Give excitation to the field winding of the motor(for separately excited motor). Set the firing angle to achieve low armature current during starting. vary the firing angle to apply different armature voltages. Observe the speed and toruqe variations. Repeat the

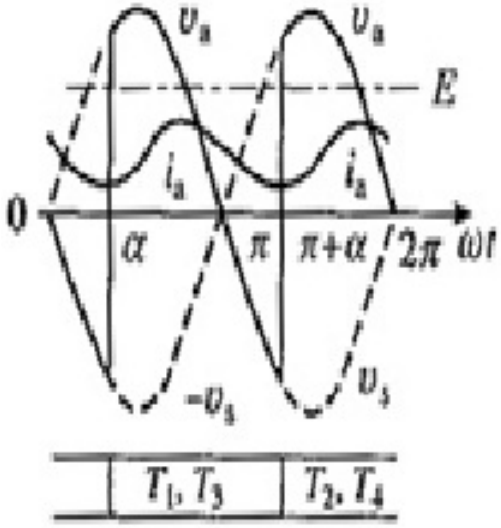


Figure 10.4: Single-phase fully-controlled converter drive output wave form in continuous conduction mode

procedure and obseve the output change with firing angle variation. Repeat the experiment after connecting the free wheeling diode.

10.4 Result

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10.5 Simulation Result

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10.6 Inference

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10.7 Outcome

After doing this experiment, the student will be able to operate and control a DC motor using a fully controlled rectifier for various appications. Analyse the continuous and discontinuous conduction modes of operation. Also understand the role of free wheeling diode.

10.8 Questions

- What is the difference in voltage wave forms with and without freewheeling diode?
- What is the effect of freewheeling diode at the output voltage?
- What is the difference between continuous and discontinuous conduction modes?